

BLUESTOCK MUTUAL FUND ANALYTICS PROJECT

End-to-End Mutual Fund Analytics, Risk Assessment, Investor Insights & Dashboard
Development

Submitted By : Bindu

Submitted To : Bluestock Fintech

Technology Stack

Programming & Analytics: Python, Pandas, NumPy, SciPy

Data Visualization: Matplotlib, Seaborn, Plotly

Database Technologies: SQLite, SQLAlchemy, SQL

Business Intelligence: Power BI

Data Collection & APIs: Requests, MFAPI

Development Tools: Jupyter Notebook, Git, GitHub

Date : 11 June 2026

EXECUTIVE SUMMARY

The Bluestock Mutual Fund Analytics Project was developed to build a comprehensive analytics platform for evaluating mutual fund performance, investor behavior, portfolio concentration, and investment risk. The project integrates multiple datasets related to mutual funds, investors, SIP inflows, benchmark indices, and portfolio holdings to generate actionable business intelligence.

The project follows an end-to-end analytics workflow beginning with data ingestion, data cleaning, validation, and database design. Advanced analytical techniques such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha-Beta Analysis, Value at Risk (VaR), Conditional Value at Risk (CVaR), Rolling Sharpe Ratio, Cohort Analysis, SIP Continuity Analysis, and Portfolio Concentration Measurement using the Herfindahl-Hirschman Index (HHI) were implemented to assess fund performance and investor engagement.

A Power BI dashboard was developed to provide interactive visualization of industry trends, fund performance, investor demographics, SIP behavior, and portfolio characteristics. The final solution demonstrates how data analytics can support investment decision-making, risk management, and investor retention strategies.

INTRODUCTION

Mutual funds have become one of the most popular investment vehicles due to their diversification benefits, professional management, and accessibility. As the mutual fund industry continues to expand, there is an increasing need for data-driven insights that can help investors evaluate fund performance, understand risks, and make informed investment decisions.

This project aims to transform raw mutual fund datasets into meaningful business intelligence through data engineering, analytics, visualization, and dashboard development.

PROBLEM STATEMENT

Investors and fund managers require deeper insights into fund performance, portfolio diversification, investment risk, and investor behavior. Traditional reporting methods often provide limited visibility into risk-adjusted returns and long-term performance trends.

The challenge is to develop a centralized analytics platform capable of processing large-scale mutual fund datasets and generating actionable insights through advanced analytics and interactive dashboards.

PROJECT OBJECTIVES

The primary objectives of the project are:

1. Build a robust ETL pipeline for mutual fund datasets.
2. Design and implement a SQLite data warehouse.
3. Perform exploratory data analysis on industry and fund-level data.
4. Measure fund performance using CAGR, Sharpe Ratio, Sortino Ratio, Alpha etc.
5. Analyze investor behavior using cohort and SIP continuity analysis.
6. Measure downside risk using VaR and CVaR.
7. Develop a recommendation engine based on investor risk appetite.
8. Create an interactive Power BI dashboard for business intelligence reporting.

DATA SOURCES

The project uses ten datasets covering mutual fund performance, investor behavior, portfolio composition, and industry trends.

Dataset	Description
01_fund_master	Fund metadata and scheme details
02_nav_history	Historical NAV records
03_aum_by_fund_house	Assets under management data
04_monthly_sip_inflows	SIP contribution statistics
05_category_inflows	Category-wise fund inflows
06_industry_folio_count	Industry folio counts
07_scheme_performance	Performance metrics
08_investor_transactions	Investor transaction history
09_portfolio_holdings	Portfolio allocation details
10_benchmark_indices	Benchmark performance data

Analysis Period: 2022–2026

Number of Schemes Analyzed: 40

ETL PIPELINE DESIGN

The project follows a structured Extract, Transform, and Load (ETL) architecture.

Extract

Raw datasets were loaded from CSV files using Pandas. Live NAV information was also fetched using MFAPI.

Transform

The transformation stage included:

- Date conversion and formatting
- Missing value handling
- Duplicate removal
- Data validation checks
- Derived metric creation
- Risk metric calculation

Load

The processed datasets were loaded into a SQLite database for analytics and reporting.

ETL Workflow:

Raw Data → Cleaning → Transformation → SQLite Database → Analytics Layer → Dashboard

DATA QUALITY AND VALIDATION

Several validation checks were performed to ensure data accuracy and reliability.

Validation Check	Result
Duplicate NAV records	Removed
Missing NAV values	Forward filled
Invalid Transactions	Corrected
AMFI code validation	Passed

Expense Ratio Validation	Passed
Data Format Validation	Passed

All AMFI scheme codes were successfully validated against NAV history records.

DATABASE DESIGN

A star-schema database architecture was implemented using SQLite.

Dimension Tables:

- dim_fund
- dim_date

Fact Tables:

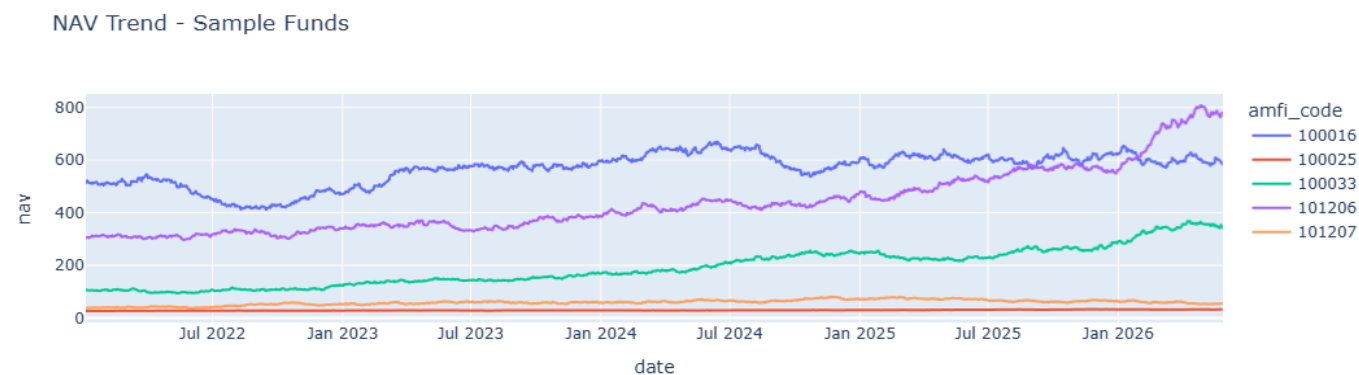
- fact_nav
- fact_transactions
- fact_performance
- fact_aum

This design improves analytical query performance and supports efficient reporting.

EXPLORATORY DATA ANALYSIS

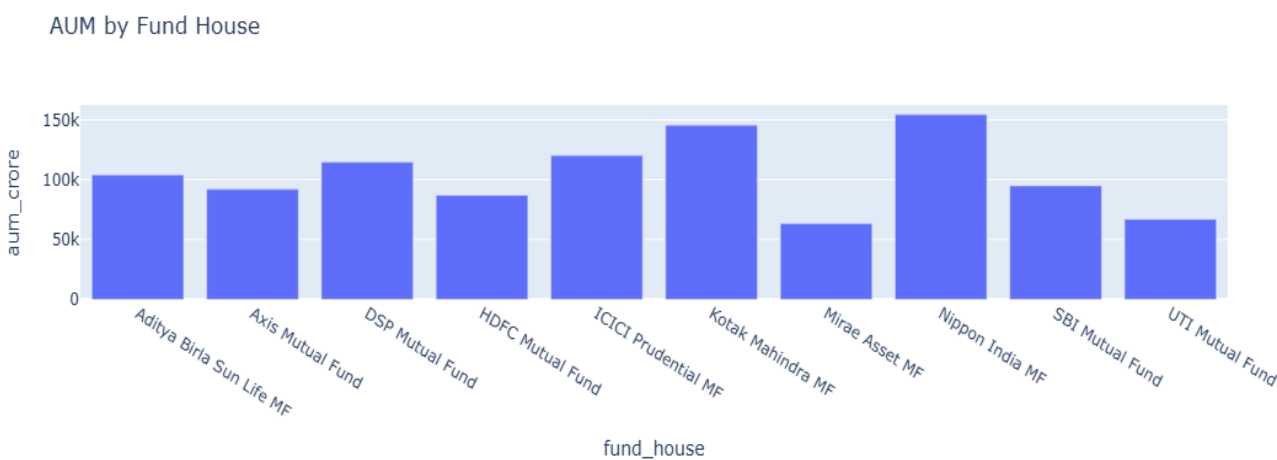
EDA was conducted to identify industry trends, investor participation patterns, and mutual fund growth characteristics.

NAV Trend Analysis



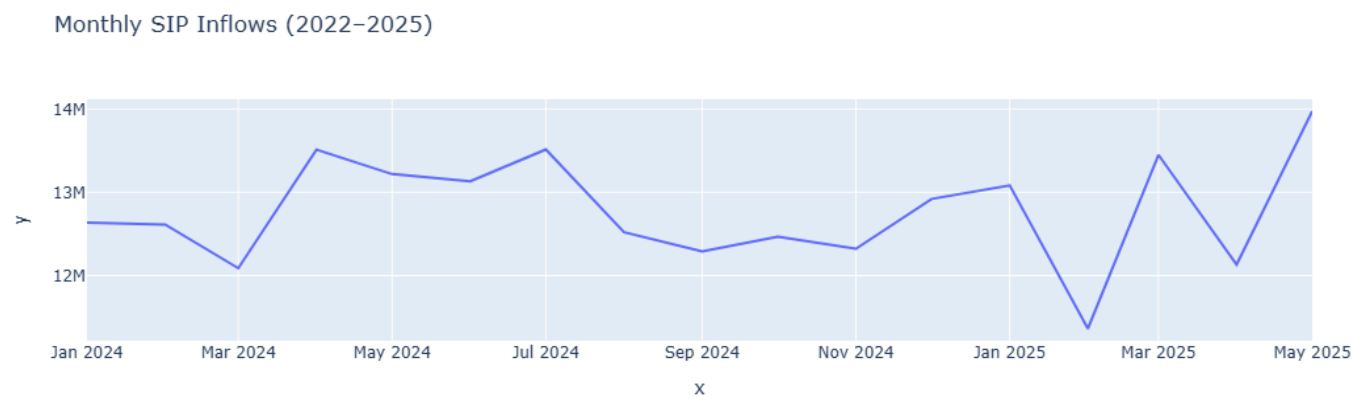
Most mutual fund schemes demonstrated long-term appreciation despite short-term market fluctuations.

AUM Growth Analysis



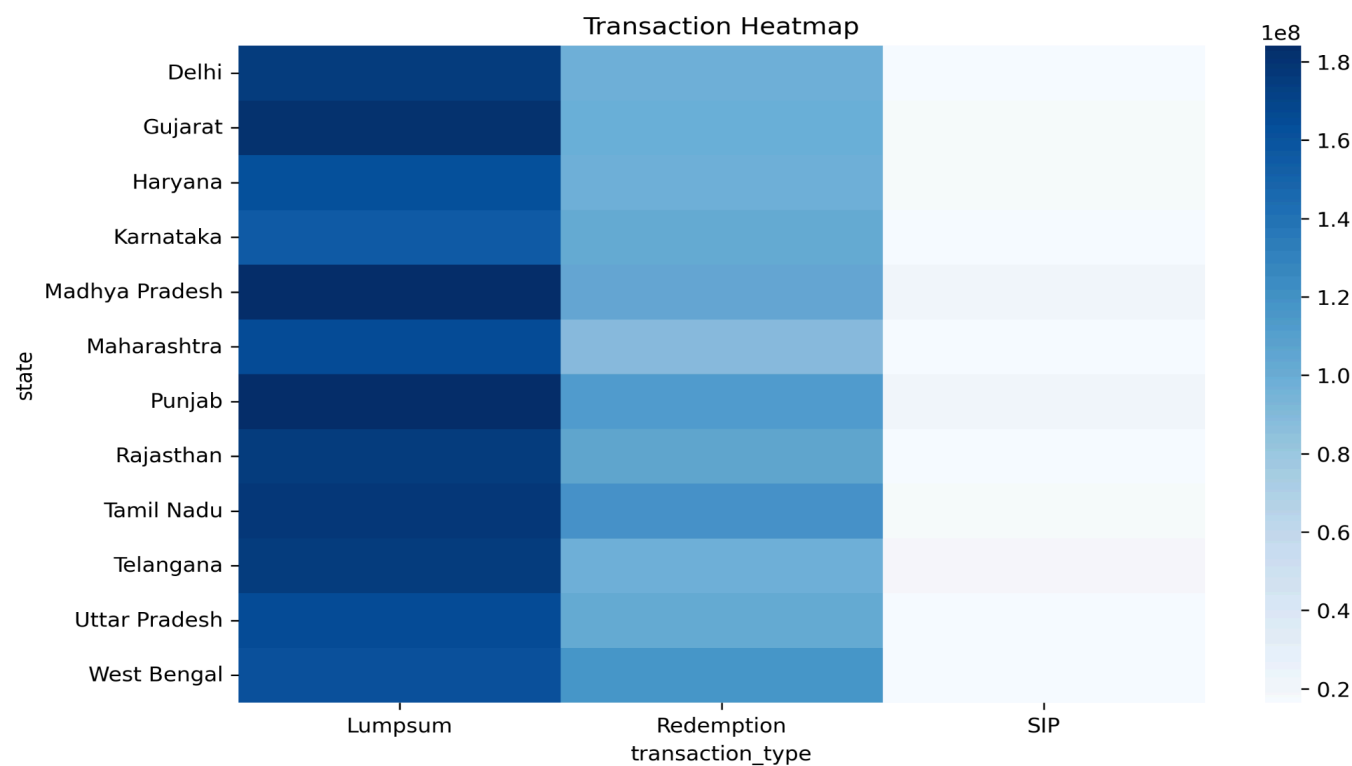
Industry assets under management showed consistent growth during the analysis period.

SIP Inflow Analysis



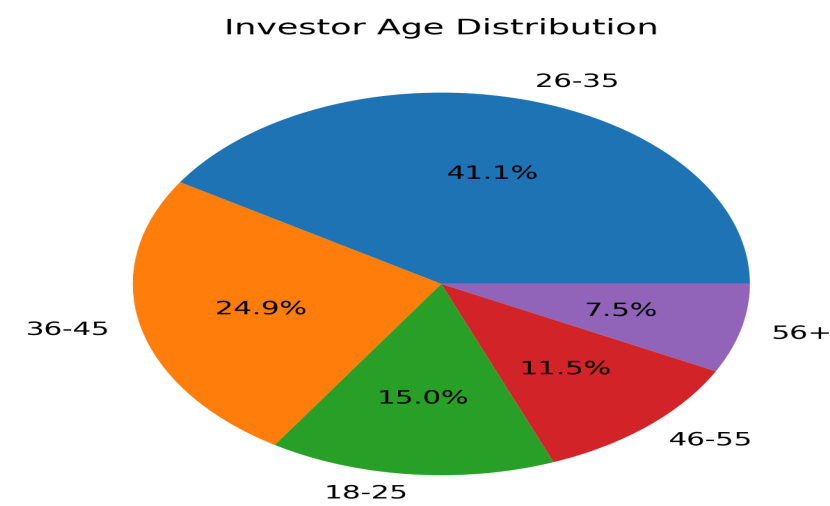
Monthly SIP inflows increased steadily and reached record highs by the end of the study period.

Category Inflow Analysis



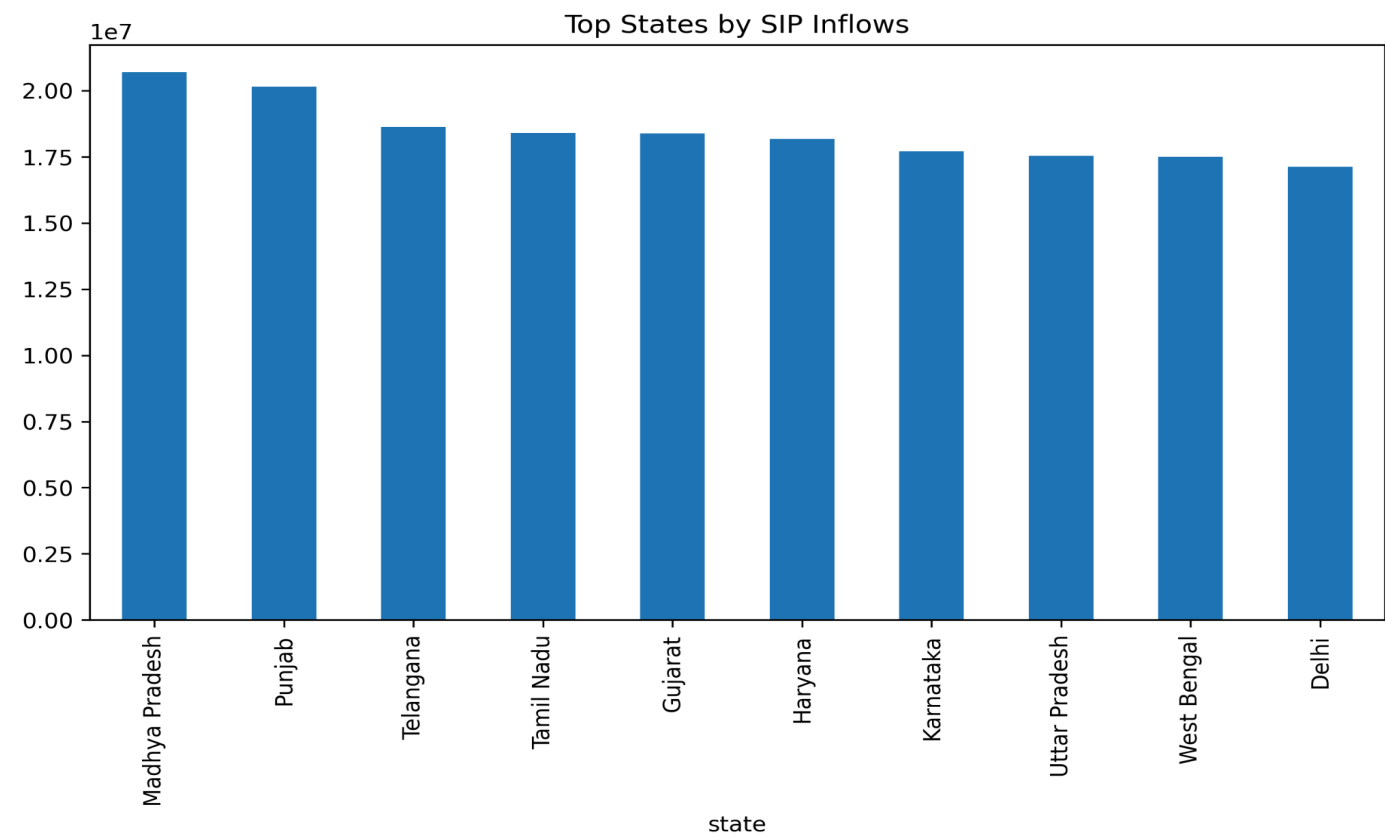
Equity-oriented categories attracted the highest investor inflows.

Investor Demographics



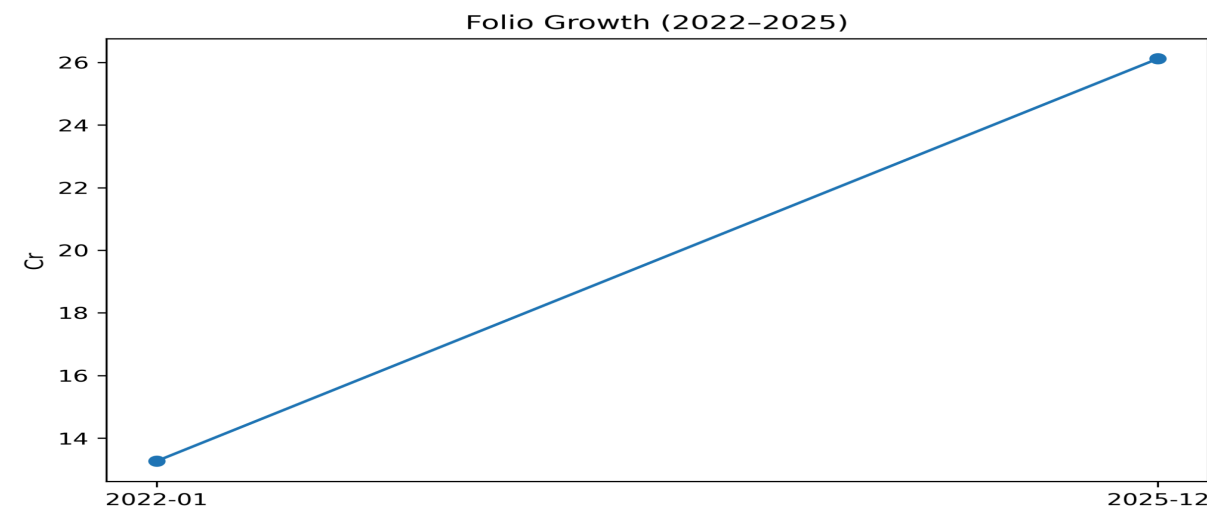
Middle-aged investors represented the largest contributor segment.

Geographic Distribution



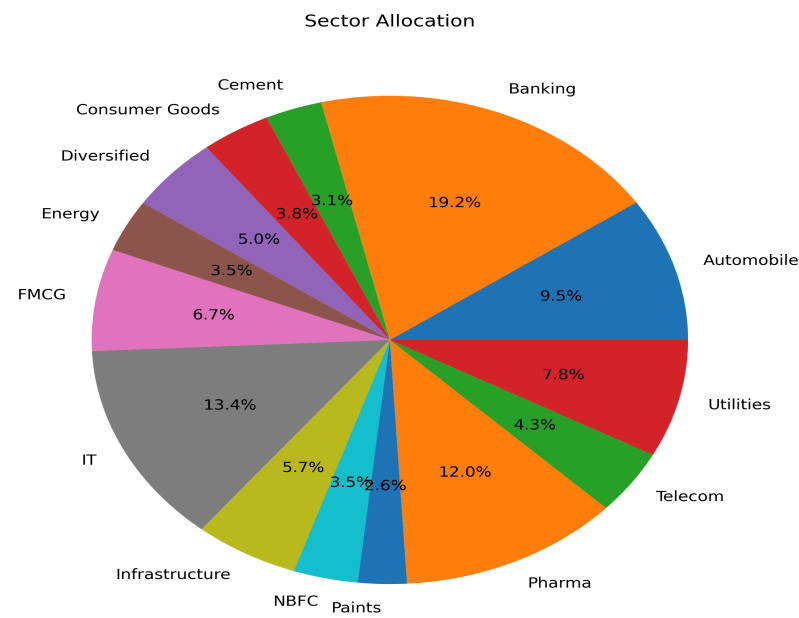
Metro cities generated the highest transaction volumes, while B30 participation continued to grow.

Folio Count Growth



Retail investor participation increased significantly over the study period.

Sector Allocation



Financial Services, Technology, and Energy sectors accounted for major portfolio allocations.

PERFORMANCE ANALYTICS

Several performance metrics were computed to evaluate mutual fund performance.

CAGR Analysis

Compound Annual Growth Rate (CAGR) was calculated for 1-year, 3-year, and 5-year periods.

Formula:

$$\text{CAGR} = (\text{Ending NAV} / \text{Beginning NAV})^{(1/n)} - 1$$

Sharpe Ratio

The Sharpe Ratio measures risk-adjusted returns.

Risk-Free Rate Used: 6.5%

Sortino Ratio

Sortino Ratio focuses on downside volatility and penalizes only negative returns.

Alpha and Beta

Regression analysis against benchmark returns was used to calculate Alpha and Beta values.

Maximum Drawdown

Maximum Drawdown identified the largest peak-to-trough decline experienced by each fund.

Fund Scorecard

A composite score was developed using:

- 30% 3-Year Return Rank
- 25% Sharpe Ratio Rank
- 20% Alpha Rank
- 15% Expense Ratio Rank
- 10% Maximum Drawdown Rank

ADVANCED ANALYTICS

Value at Risk (VaR)

Historical VaR at the 95% confidence level was calculated using daily return distributions.

Conditional Value at Risk (CVaR)

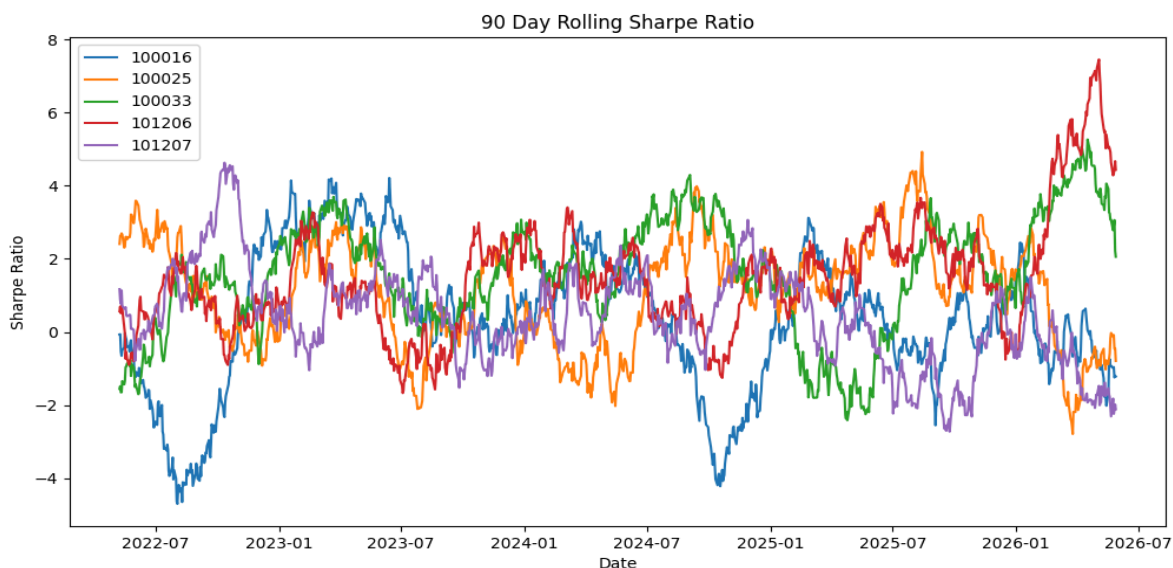
CVaR was calculated as the average loss beyond the VaR threshold.

Fund 101207 exhibited the highest downside risk with:

VaR = -2.60%

CVaR = -3.25%

Rolling Sharpe Ratio



Rolling 90-day Sharpe Ratios were analyzed to evaluate consistency in risk-adjusted performance.

Investor Cohort Analysis

Investor cohorts were created based on first transaction year.

The 2025 cohort demonstrated slightly higher average investment levels compared to earlier cohorts.

SIP Continuity Analysis

A total of 1,332 investors were identified as at-risk due to SIP gaps exceeding 35 days.

Fund Recommendation Engine

A rule-based recommendation engine was developed to recommend the top three funds based on investor risk appetite and Sharpe Ratio rankings.

Portfolio Concentration Analysis

Sector concentration was measured using the Herfindahl-Hirschman Index (HHI).

Fund 119092 recorded the highest concentration score, indicating greater sector exposure risk.

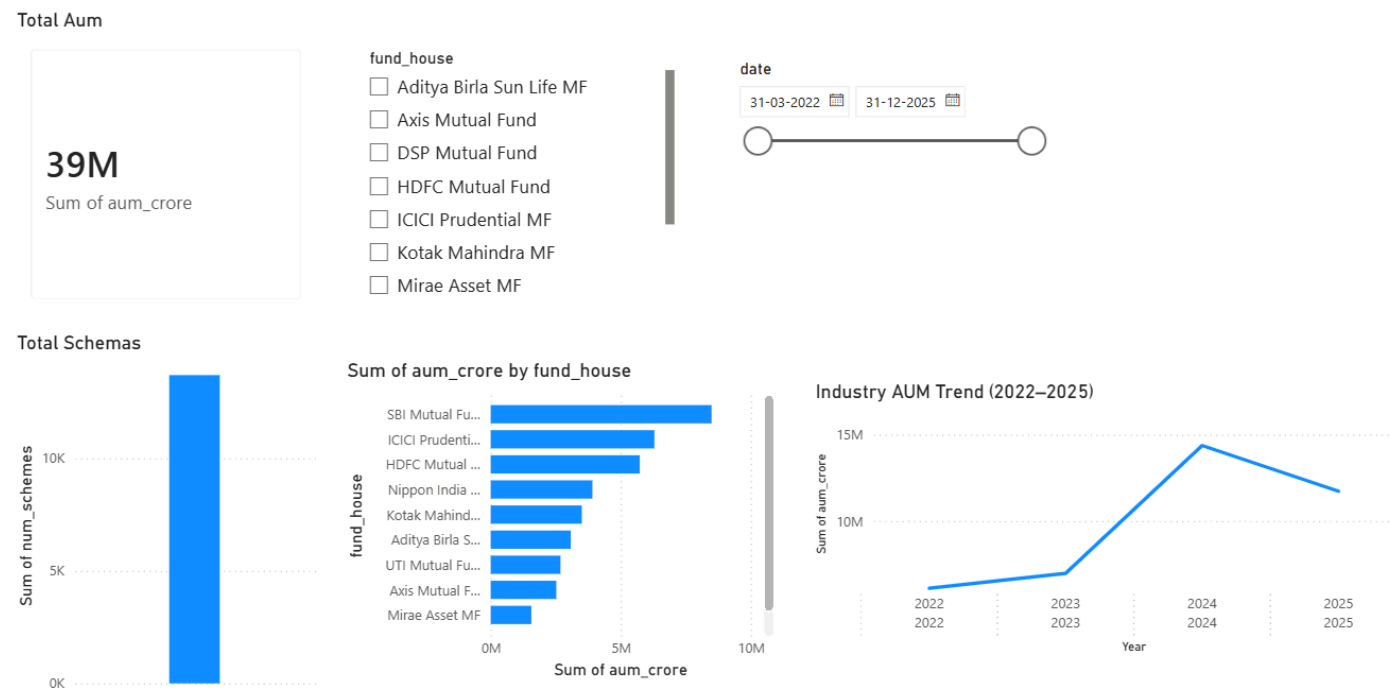
POWER BI DASHBOARD

A four-page Power BI dashboard was developed.

Page 1: Industry Overview

Features:

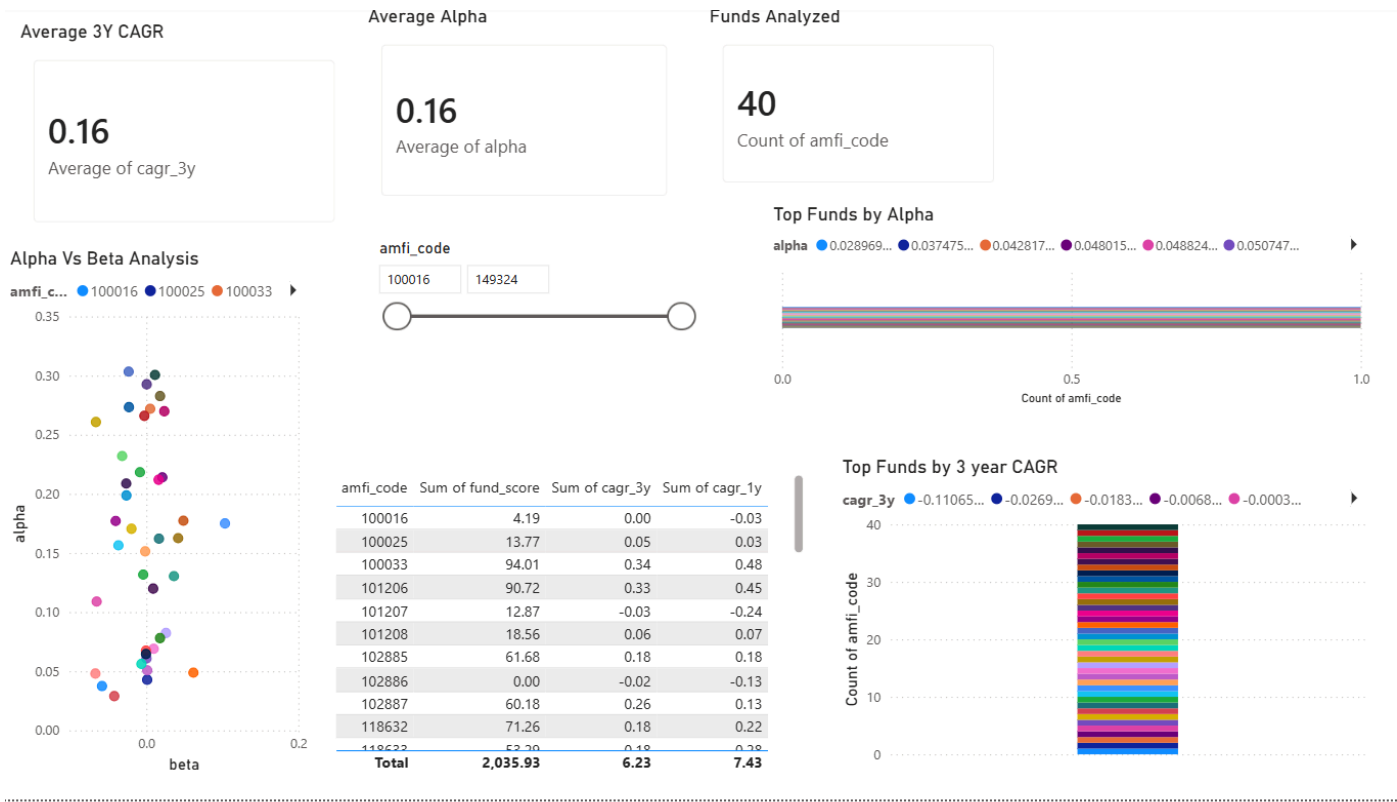
- Total AUM
- SIP Inflows
- Folio Counts
- Industry Growth Trends



Page 2: Fund Performance

Features:

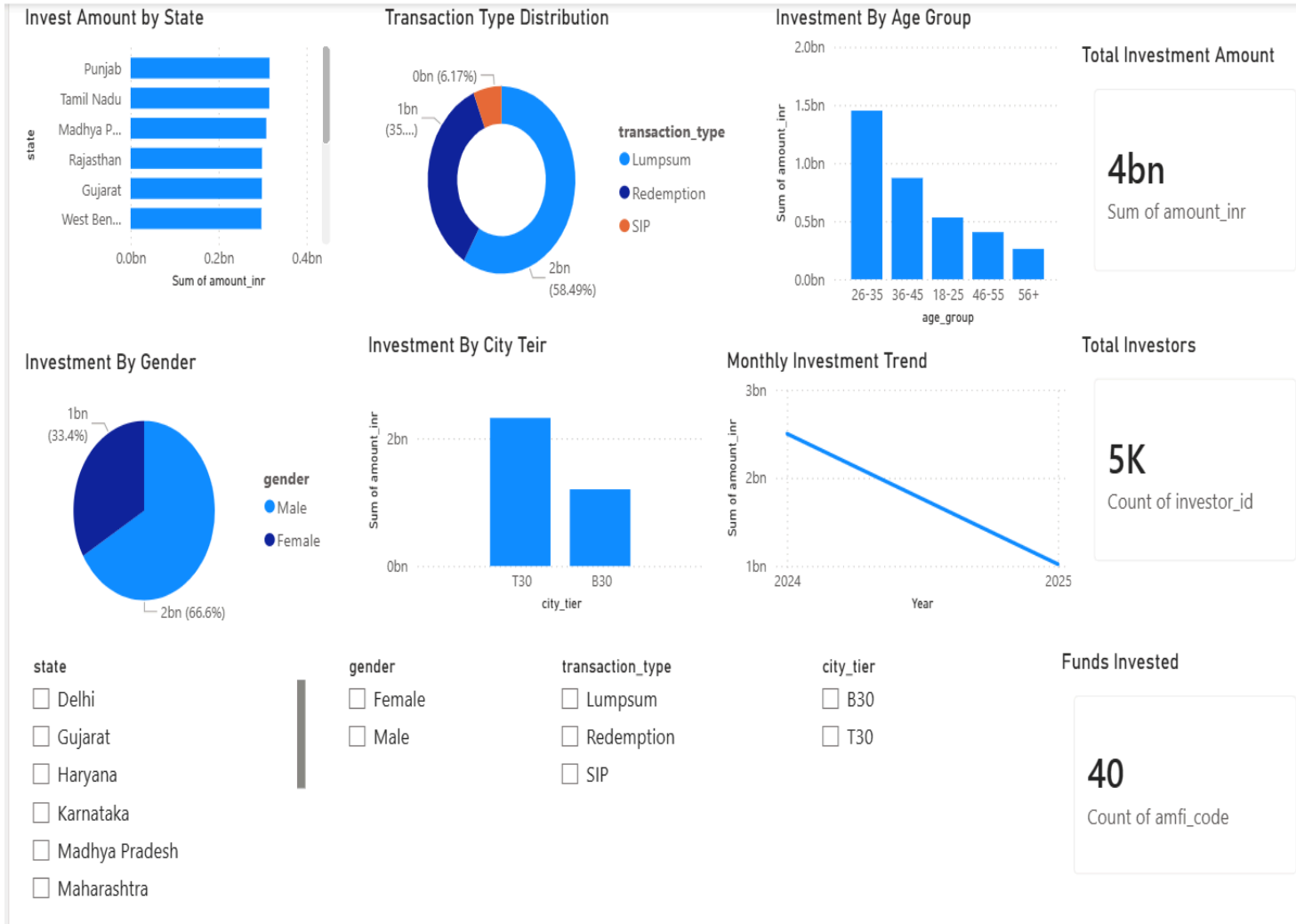
- Risk vs Return Analysis
- Fund Scorecards
- NAV Comparison



Page 3: Investor Analytics

Features:

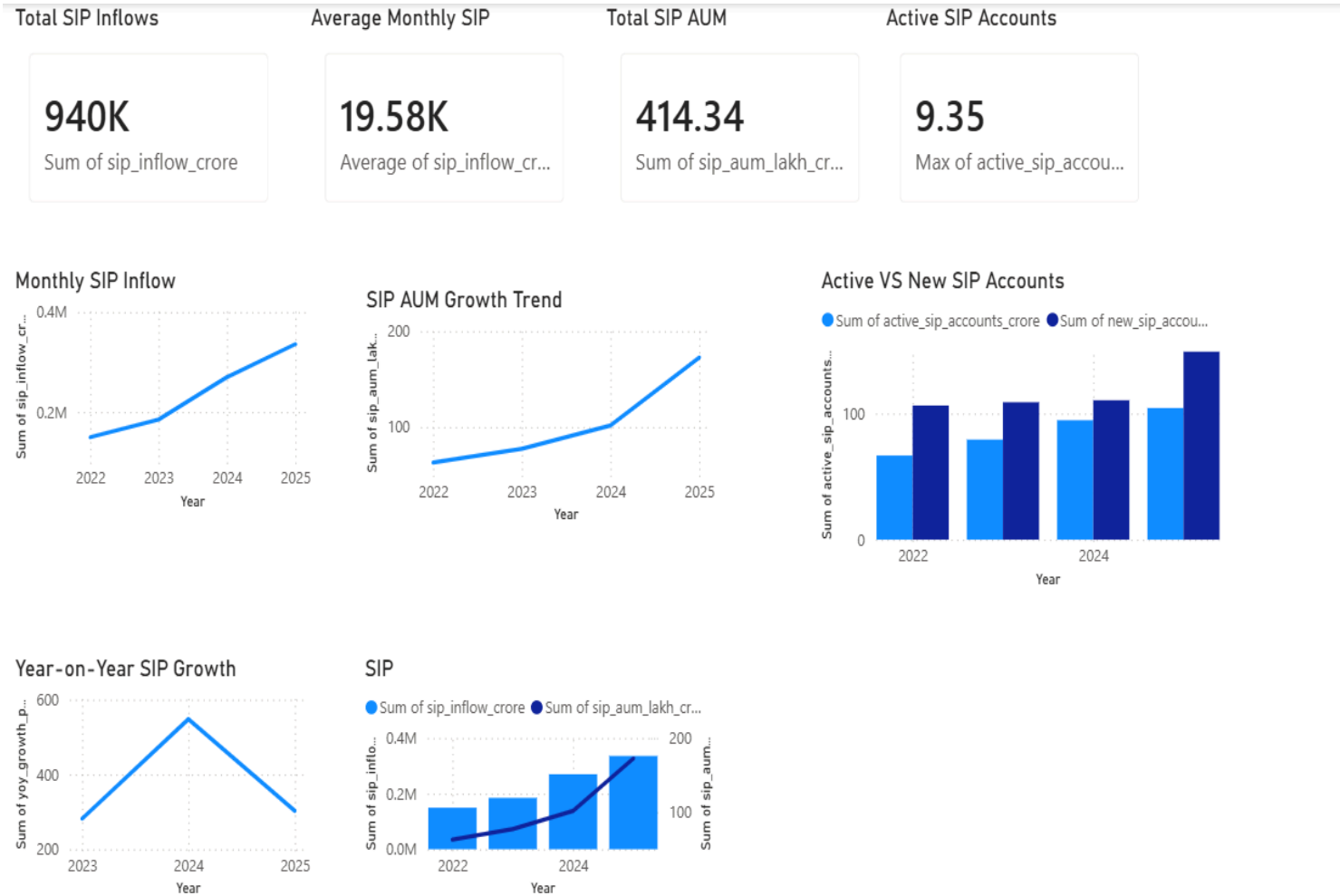
- State-wise Investments
- Demographic Analysis
- Transaction Trends



Page 4: SIP & Market Trends

Features:

- SIP Inflow Trends
- Category Analysis
- Market Comparison



KEY FINDINGS

1. Fund 101207 exhibited the highest downside risk among all analyzed schemes.
2. Investor participation increased consistently during the study period.
3. SIP inflows demonstrated strong growth and reached record highs.
4. Equity-oriented categories attracted the largest inflows.
5. Large-cap schemes showed strong positive return correlations.
6. The 2025 investor cohort invested more on average than earlier cohorts.
7. A significant number of investors displayed SIP discontinuity risk.
8. Portfolio diversification varied substantially across schemes.
9. Certain funds demonstrated superior risk-adjusted performance.
10. Dashboard analytics improved visibility into industry and investor trends.

LIMITATIONS

- Historical datasets were used for analysis.
- Real-time portfolio rebalancing was not included.
- Recommendation engine used rule-based logic.
- Benchmark universe was limited.

RECOMMENDATIONS

1. Integrate real-time NAV feeds for continuous monitoring.
2. Develop machine learning-based recommendation models.
3. Implement predictive SIP churn analysis.
4. Expand benchmark coverage.
5. Deploy dashboards through cloud-based reporting solutions.

CONCLUSION

The Bluestock Mutual Fund Analytics Project successfully demonstrated the application of data engineering, financial analytics, and business intelligence techniques to mutual fund datasets. Through ETL pipelines, database design, exploratory data analysis, performance analytics, and advanced risk modeling, the project generated valuable insights into fund performance and investor behavior.

The Power BI dashboard provides an interactive platform for monitoring industry trends, evaluating fund performance, and understanding investor engagement. The project highlights how data-driven decision-making can support both investors and fund managers in achieving better investment outcomes.